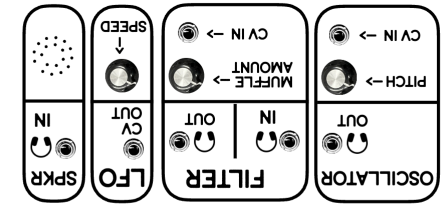
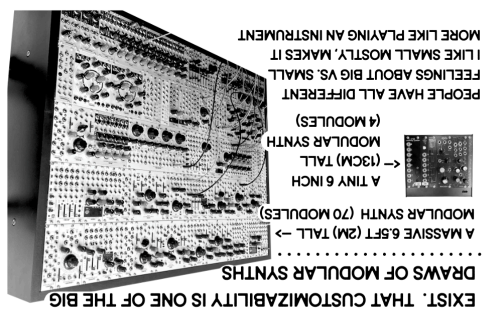


* WHAT MAKES A SYNTH *

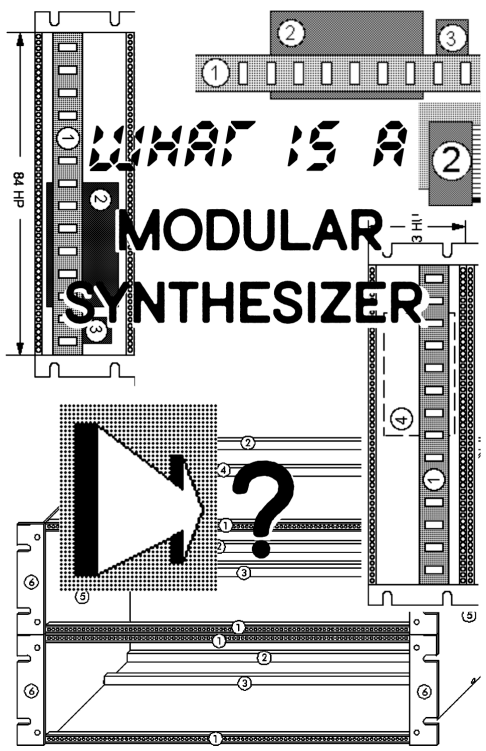
A TYPICAL SYNTHESIZER IS MADE UP OF MULTIPLE DIFFERENT SUB-SYSTEMS, WHICH ARE WIRED TOGETHER IN A BOX. USUALLY WITH A MUSICAL KEYBOARD. THE MANUFACTURER PICKS THE COLLECTION OF SUB SYSTEMS AND SAYS, THIS WOULD MAKE A GOOD COMPLETE SET OF SUB-SYSTEMS, AND SELLS IT. LET'S IMAGINE ONE:

1. OSCILLATOR: MAIN SOUND SOURCE, BUZZY SOUNDING, PITCHED CONTROLLED DIRECTLY BY KEYBOARD
2. FILTER: GRADUALLY MUFFLES THE BUZZY OSCILLATOR BY TURNING A KNOB
3. ENVELOPE: ACTIVATED WHENEVER A KEYBOARD KEY IS PRESSED. AUTOMATICALLY CONTROLS THE VOLUME TO CHANGE NOTE 'SHAPE' (HOW FAST DOES THE VOLUME FADE IN WHEN YOU PRESS A KEY, HOW FAST DOES IT FADE OUT WHEN YOU RELEASE ONE? DOES A NOTE PLAY FOREVER WHEN HELD DOWN, OR IS IT A SHORT BLIP?) YOU DIAL THAT IN WITH A FEW KNOBS
4. LFO: AUTOMATICALLY AND CONSTANTLY TURNS SOMETHING UP AND DOWN CYCLING, USUALLY PITCH OR FILTER LEVEL. YOU CONTROL SPEED/INTENSITY



PATCHABILITY. REMEMBER, ON A NORMAL SYNTH, THE ELEMENTS ARE DIRECTLY WIRED TO EACH OTHER, IN A MODULAR SYNTH, EVERYTHING IS MANUALLY WIRED/REWIRED ON THE FLY, TEMPORARILY AND RECONFIGURABLY - THE IS DONE WITH PATCH CABLES. WHICH LOOK LIKE AN AUX CABLE, BUT CAN PASS EITHER AUDIO OR INFORMATION (CV). MODULES HAVE 'INPUT' AND 'OUTPUT' JACKS (THE HOLES LIKE AN AUX CABLE GOES INTO). LET'S IMAGINE A SYSTEM THAT HAS ONE OSCILLATOR, ONE FILTER, ONE LFO, AND A SPEAKER. WIRING THE AUDIO IS SIMPLE, WE TAKE A CABLE AND CONNECT WHERE THE SOUND IS COMING OUT OF THE OSCILLATOR (THE AUDIO 'OUTPUT' JACK) AND PLUG THAT INTO THE FILTER AUDIO 'INPUT' JACK. THEN PLUG THE AUDIO 'OUTPUT' JACK OF THE FILTER INTO THE SPEAKER. THE FILTER TAKES AUDIO INTO THE SPEAKER, THEN SENDS IT OUT. IN, MUFFLES IT, THEN SENDS IT OUT.

WE CAN CONTROL THE INTENSITY OF THE MUFFLING WITH A KNOB ON THE FILTER, BUT BELOW THAT KNOB IS A SPECIAL JACK CALLED 'CONTROL VOLTAGE (CV)' IN WHICH LETS US CHANGE WHATEVER THAT KNOB NORMALLY CHANGES, BY PLUGGING SOMETHING INTO IT THAT SENDS CV. OUR LFO 'OUT' JACK DOESN'T SEND AUDIO, BUT RATHER CV, CONSTANTLY TURNING UP AND DOWN, SO IF WE PLUG THE LFO INTO THE FILTER CV IN, IT WILL CONTROL THE FILTER MUFFLE, BUT WE CAN CHOOSE TO PLUG IT IN ELSEWHERE. THE OSCILLATOR HAS A CV INPUT FOR PITCH, IF WE WIRE THE LFO OUT TO IT, IT WILL SLOWLY TURN THE PITCH UP AND DOWN, IMAGINE WE HAVE A REVERB MODULE INSTEAD OF A FILTER - WE COULD USE THE LFO TO TURN THE REVERB AMOUNT UP AND DOWN, OR THE SIZE OF THE REVERBERATING SPACE. THE LFO CAN EFFECT ANYTHING THAT ACCEPTS A CV IN, AND SIMILARLY, WE CAN USE ALL SORTS OF DIFFERENT LFO-LIKE MODULES AS CV MAKERS: INSTEAD OF SIMPLY GOING UP AND DOWN, ONE MIGHT HAVE A COMPLICATED LOOPING PATTERN, OR A RANDOM PATTERN, OR A PATTERN THAT YOU DESIGN WITH CONTROLS ON THE MODULE. THIS MAKES MODULAR SYNTHS NOT ONLY INFINITELY CUSTOMIZABLE BUT INFINITELY CONFIGURABLE



WHAT NOW

EURORACK! IS INCREDIBLY EXPENSIVE ... A SIMPLE SYSTEM CAN RETAIL FOR \$1500+ ALL THE WAY UP TO YOUR WHOLE SAVINGS ACCOUNT. THERE'S A FEW WAYS TO DEAL WITH THIS. YOU CAN ASSEMBLE OR 3D PRINT YOUR OWN CASE, AND YOU CAN BUY AND ASSEMBLE KITS FOR MUCH CHEAPER THAN NORMAL. YOU'LL NEED TO LEARN HOW TO SOLDER, BUT IT'S EASY (CHECK OUT SITES LIKE "THONK.CO.UK" FOR KITS). THERE'S ALSO CHEAPER FORMATS LIKE "AE MODULAR". IF YOU WANT TO PLAY AROUND WITH MODULAR SYNTH CONCEPTS YOU CAN ALSO USE A VIRTUAL SIMULATOR LIKE "VCV RACK" WHICH IS FREE AND OPEN SOURCE, AND HAS LOTS OF CUSTOM MODULES MADE BY PEOPLE. IF YOU WANT TO CHECK OUT A BUNCH OF REAL EURORACK MODULES AND CASE CONFIGURATIONS CHECK OUT "MODULARGRID.NET". THERE'S ALSO 'SEMI MODULAR' SYNTHS WHICH ARE LIKE A PREMADE COLLECTION OF CUSTOM PATCHABLE MODULES - "VOLCA MODULAR" AND "BASTL KASTLE 2" ARE GREAT CHEAP ONES

* WHAT IS "EURORACK" *

EURORACK IS A SPECIFIC TYPE OF MODULAR SYNTHESIZER, A 'FORMAT'. BASICALLY, IN 1995 THE GERMAN SYNTH COMPANY DOEPFER CAME OUT WITH A SET OF 10 SYNTH MODULES IN A BOX, CALLED THE 'A-100' SYSTEM. THAT SYSTEM USES A SPECIFIC SET OF TECHNICAL STANDARDS - STUFF LIKE, WHAT VOLTAGE SHOULD WE POWER THE MODULES WITH? WHAT KIND OF CABLES DO WE CONNECT THEM WITH? HOW BIG SHOULD THE CASE AND MODULES BE? RATHER THAN KEEPING ALL OF THAT SECRET, THEY PUBLICLY RELEASED THE DETAILS OF THE SYSTEM SO THAT ANYONE COULD DESIGN/SELL THEIR OWN MODULES AND THEY WOULD FIT IN THE BOX, AND SAFELY INTERACT WITH ANY OTHER MODULES. THAT STANDARD IS CALLED "EURORACK". THERE ARE NOW THOUSANDS OF EURORACK MODULES, MAKE BY ALL DIFFERENT COMPANIES, INDIVIDUALS, ETC. AND THEY ARE ALL COMPATIBLE WITH EACH OTHER BECAUSE THEY FOLLOW THAT SAME TECHNICAL SET OF RULES ... THERE ARE A HANDFUL OF OTHER MODULAR FORMATS, BUT EURORACK IS THE MOST POPULAR. IF SOMEONE IS TALKING ABOUT 'MODULAR' THEY'RE LIKELY TALKING ABOUT EURORACK